Syllabus for APEC 3611w: Environmental and Natural Resource Economics

University of Minnesota
Department of Applied Economics

Spring 2026, Mon/Wed/Fri, 10:40 to 11:30, Ruttan Hall, B25

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3 Credit course.

Course website: https://justinandrewjohnson.com/teaching/apec_3611
Course Canvas site: https://canvas.umn.edu/courses/540318

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Course Content and Objectives

This course introduces upper-level undergraduates to the economic analysis of natural resources and the environment, framed through the lens of *earth–economy systems*. We study how markets, institutions, ecosystems, and technologies interact to shape resource use and environmental outcomes. Traditional topics—externalities, public goods, market failure, valuation, cost–benefit analysis, renewable and non-renewable resources—are integrated into a coherent systems view in which the economy is embedded within the biosphere.

Students learn to reason across scales, from individual decisions to global dynamics, and to connect microeconomic tools with macro-level outcomes such as climate change, biodiversity loss, and sustainable development. The course emphasizes *earth–economy modeling*: simplified representations of coupled human–natural systems used to explore tradeoffs, policies, and futures.

Learning Objectives

By the end of the course, students will be able to:

1. Explain why environmental problems arise in market economies and how they relate to externalities, public goods, and the commons.
2. Analyze resource use decisions for renewable and non-renewable resources using economic models.
3. Measure and value environmental impacts using economic tools.
4. Evaluate environmental policies using efficiency, equity, and sustainability criteria.
5. Understand the economy as embedded within ecological systems.
6. Build and interpret simple *earth–economy models* that link economic activity with environmental change.
7. Use models to explore tradeoffs, uncertainty, and long-run dynamics in environmental policy.

Class Hours and Mode of Instruction

This is an IN-PERSON class! We will meet for lectures and discussion section at the times and locations listed below. Attendance is required and 10% of your grade will depend on your attendance and participation. Additionally, classes will include worktime on exercises along with hints on how to complete them (so you will definitely want to come!). However, if you have an excused absence (discussed below), you will not lose any points. Email me in advance (ideally) to confirm.

Every lecture is recorded and posted publicly, embedded on that lecture's page and collected in the course YouTube playlist linked from the home page. The camera stays on me and never points at the class, so student images are not captured, but student voices may be. If you would rather not be recorded at all, talk to me before the recordings go up. Recordings are a reference, not a substitute for attending: attendance is still part of your grade.

A detailed schedule is on the course website, and Canvas links to it.

Course Textbook and Readings

The primary textbook for this course is:

Keohane, Nathan and Sheila Olmstead, Markets and the Environment, second edition, (Washington DC: Island Press, 2016), available for free by UMN libraries at Markets and the Environment (You need to be logged in with your UMN credentials to access the full text.)

Course Website

This course embraces open-access publishing (and subsequent years will likely switch to a new open-access textbook to be created by me!). For now, all course materials, including lecture slides, problem sets, and additional readings, will be available on the course website at https://justinandrewjohnson.com/teaching/apec_3611. Canvas https://canvas.umn.edu/courses/540318 will be used minimally, only for announcements, assignment submission and grade reporting.

Course Schedule

The most up-to-date course schedule is found on the Course Homepage. Below, I reproduce the schedule as of the start of the semester for reference.

Date	Title	Subtitle
1/21/2026	Introduction	Environmental and Natural Resource Economics in a Global Framing
1/23/2026	The Doughnut	Macroeconomic Feasibility
1/26/2026	The Microfilling	Microeconomic Decision-making
1/28/2026	Supply and Demand	The power of equilibrium and Micro-Macro Synthesis
1/30/2026	Cost benefit analysis	The standard workhorse of environmental economics
2/2/2026	Optimal Pollution	A contradiction?
2/4/2026	Externalities	Spillovers
2/6/2026	Public Goods	Underprovision is likely
2/9/2026	Commons	Always a Tragedy?
2/11/2026	The Whole Economy	Partial vs General Equilibrium
2/13/2026	GDP and the Environment	Measuring what matters
2/16/2026	Kuznet's Curve	Does development improve the environment?
2/18/2026	Inclusive Wealth	A better metric?

Date	Title	Subtitle
2/20/2026	Dynamic Efficiency and Sustainable Development	Methods to define sustainability
2/23/2026	Climate Change	The ultimate externality
2/25/2026	Social Cost of Carbon	Manage per ton of carbon
2/27/2026	DICE and Climate Skepticism	It all depends on the social cost of carbon
3/2/2026	Air Pollution	Spillovers in the air
3/4/2026	Water Pollution	Spillovers on the ground
3/6/2026	MIDTERM EXAM	
3/16/2026	Non-renewable resources 1	Will we run out?
3/18/2026	Non-renewable resources 2	How much should we extract?
3/20/2026	Renewable resources 1	Fisheries
3/23/2026	Renewable resources 2	Forestry
3/25/2026	Land as a resource	The forgotten input
3/27/2026	Land-use change prediction	Tools to predict where nature is at risk
3/30/2026	Ecosystem Services and Natural Capital	Return to the Micro-Macro relationship
4/1/2026	Valuing Nature	Bottom-up methods for putting a monetary value on nature
4/3/2026	Biodiversity	Species protection and the fundamental source of value
4/6/2026	Ecosystem services 1	Introduction and context
4/8/2026	Ecosystem services 2	Methods and specific services
4/10/2026	Ecosystem Services 3	More services and policy directions
4/13/2026	Risk, Uncertainty and Tipping Points	Regime change and hysteresis
4/15/2026	Scenarios Exploring Possible Futures	Shared socio-economic pathways
4/17/2026	Positive Visions of the Future	Nature Futures Framework
4/20/2026	Designing policies for earth-economy systems	Global policies and cooperation
4/22/2026	Market-based policies	Micro-level optimization
4/24/2026	Environmental policies in the real world	Quotas, Trading and Taxes
4/27/2026	Toy Earth-Economy models	A video game for sustainability?
4/29/2026	Gridded Earth-Economy models	Millions of markets
5/1/2026	Earth-economy modeling in practice	GTAP-InVEST
5/4/2026	Conclusion	Where to go from here?
5/12/2026	Final Exam: 8:00-10:00 AM	

Reminder, see the full and up-to-date schedule on the Course Homepage.

Prerequisites

APEC 1101 or equivalent (undergraduate level principles of microeconomics). It is possible to take microeconomics concurrently. If you haven't had such classes (or are taking them concurrently) please talk to me about what you will need to do to keep up.

Course Grading

Class participation 10%
Weekly questions 20%
Assignments 10%
Quizzes 10%
Midterm 15%
Writing Assignment(s) 5%

Final project 15%
Final exam 15%

Letter Grades

Class Points

Class participation is scored partly through class points (CP), an incentive system rather than a penalty. Doing the ordinary work of showing up and taking part will generally earn the full 10%; class points are on top of that.

Points are awarded on the spot for things like making a good point in discussion, and through in-class activities such as the game theory demonstrations. They are spendable: 5 CP waive the late penalty on any assignment, and 10 CP earn full credit on a weekly question. Points are never taken away, and whoever finishes the semester with the most has earned some bragging rights.

Assignments and In-Class Quizzes

Assignments are done at home with whatever tools you like, including AI. On the day each one is due there is a short in-class quiz covering the same problems with different numbers. If you worked through the assignment yourself the quiz is quick; if you had something else do it for you, it will not be. The pairing keeps the take-home work useful for learning while still checking that the learning happened.

Letter grades will be assigned using the scale below. The usual rounding rules apply.

A: 100 ~ 93
A-: 92 ~ 90
B+: 89 ~ 87
B: 86 ~ 83
B-: 82 ~ 80
C+: 79 ~ 77
C: 76 ~ 73
C-: 72 ~ 70
D+: 69 ~ 67
D: 66 ~ 60
F: 59 ~ 0

Late Work Policy

Submit assignments on time or arrange extensions in advance. Late work without valid excuse loses 10% of the total points per day, but not falling below 50% value. Last-minute crises due to procrastination are not acceptable excuses.

Academic Dishonesty and Plagiarism

The University of Minnesota defines academic dishonesty as "Submission of false records of academic achievement; cheating on assignments or examinations; plagiarizing; altering, forging, or misusing a University academic record; taking, acquiring, or using test materials without faculty permission; acting alone or in cooperation with another to falsify records or to obtain dishonestly grades, honors, awards, or professional endorsement" (University of Minnesota's Board of Regents Student Conduct Code). Plagiarism is the "use the words or ideas of another person as if they were your own words or ideas" (Merriam Webster Dictionary). If you want to use the exact wording from a previously published work in your own work you must put the wording in quotation marks and cite the source (as shown by example in the prior sentence). If you use ideas or specific facts from a source but do not use the exact words then you still must cite the source of the original ideas or facts. Evidence of academic dishonesty will be forwarded to the Student Scholastic Conduct Committee. TurnItIn is used to check for plagiarism on written assignments.

Credits and Workload Expectations

One credit is defined as equivalent to an average of three hours of learning effort per week (over a full semester) necessary for an average student to achieve an average grade in the course. For example, a student taking a three credit course that meets for three hours a week should expect to spend an additional six hours a week on coursework outside the classroom.

Students with Disabilities

The University of Minnesota is committed to providing equitable access to learning opportunities for all students. Disability Services (DS) is the campus office that collaborates with students who have disabilities to provide and/or arrange reasonable accommodations. If you have, or think you may have, a disability (e.g., mental health, attentional, learning, chronic health, sensory, or physical), please contact DS at 612-626-1333 to arrange a confidential discussion regarding equitable access and reasonable accommodations. If you are registered with DS and have a current letter requesting reasonable accommodations, please let me know early in the semester so we can agree on accommodations that will be applied in the course.

Students with Mental Health Issues

As a student you may experience a range of issues that can cause barriers to learning, such as strained relationships, increased anxiety, alcohol/drug problems, feeling down, difficulty concentrating, and/or lack of motivation. These mental health concerns or stressful events may lead to diminished academic performance or reduce your ability to participate in daily activities. University of Minnesota services are available to assist you with addressing these and other concerns you may be experiencing. You can learn more about the broad range of confidential mental health services available on campus via www.mentalhealth.umn.edu.

AI Usage Policy

You are free to use AI in whatever way you want, but you must attribute when you use it with a disclaimer in any assignment. I allow AI because because AI will be a critical tool in the future and so we should learn how best to use it.

Two important caveats:,

1. Mistakes made by the AI are considered YOUR mistakes and you are fully responsible for them.
2. Using AI is a double-edged sword; you will be able to complete tasks faster but it will reduce your comprehension of the topic. It increases productivity but not understanding. If you rely on AI for assignments, you will likely do worse on the in-class exams.